PRODUCT REQUIREMENTS DOCUMENT

Data Drift Intelligence System

for Microfinance Institutions

“Monitors ML loan default prediction models for data drift, alerting MFIs before bad loans spike due to economic shifts.”

Version	1.0 — Initial Release
Date	April 2026
Status	Draft — Ready for Review
Audience	Engineering, Product, Stakeholders
Domain	MLOps / Microfinance / FinTech

1. Executive Summary

One-Line Pitch: Monitors ML loan default prediction models for data drift, alerting MFIs before bad loans spike due to economic shifts.

Microfinance Institutions (MFIs) in India increasingly rely on ML models (XGBoost, LightGBM) to approve loans for 500M+ unbanked individuals. However, macroeconomic shifts — post-monsoon income drops, inflation cycles, election-driven spending — cause data drift that silently degrades these models from 82% accuracy to as low as 65%, leading to ₹15 Cr+ in annual losses per institution.

The Data Drift Intelligence System (DDIS) is an end-to-end monitoring platform that automatically detects when a deployed credit risk model's input data starts diverging from its training distribution, triggers intelligent alerts via Slack/Email, and recommends or executes retraining — before default rates spike.

2. Problem Statement

2.1 The Core Problem

ML credit scoring models are trained on a static snapshot of borrower data. Real-world economic conditions — and thus borrower profiles — evolve continuously. When the production data distribution shifts significantly from the training distribution (data drift), model predictions become unreliable.

Trigger Event	Feature Affected	Impact
Post-Monsoon (Oct-Dec)	AMT_INCOME_TOTAL, DTI Ratio	Farmers' income drops 30-40%
Inflation Spike	EXT_SOURCE scores, repayment behavior	Credit stress across all segments
Election Season	Spending patterns, loan demand	Unusual income & cash flow spikes
Fintech Onboarding Wave	EXT_SOURCE, credit history depth	New-to-credit users skew features

2.2 Business Impact of Undetected Drift

- Model accuracy degrades from 82% → 65% silently
- Default rate climbs from 14% to 18-22% before anyone notices
- ₹15 Cr+ in annual losses per mid-sized MFI

- RBI AI governance mandates go unmet, creating compliance risk
- Loan officers make decisions on stale model scores with no warning

3. Solution Overview

3.1 System Architecture

DDIS is a 4-component pipeline that wraps around any existing ML credit scoring model:

Component	What It Does	Tech
1. Baseline Model	XGBoost model trained on Home Credit dataset to predict loan default (TARGET=1)	Python, XGBoost/LightGBM, Scikit-learn
2. Drift Monitor	Compares reference (training) vs production (live) data distributions daily	Evidently AI, PSI, Chi-square tests
3. Alert Engine	Fires Slack/Email alerts when drift exceeds thresholds with context & severity	SMTP, Slack API, PostgreSQL
4. Dashboard	Visual drift scores, feature-level plots, and retraining recommendations in a React web dashboard served by FastAPI	React, TypeScript, Recharts/Plotly.js, FastAPI, PostgreSQL

3.2 End-to-End Workflow

1. TRAIN — XGBoost trained on reference data (application_train.csv, 307K rows)
2. DEPLOY — Model generates live predictions on incoming loan applications
3. MONITOR — Daily batch job compares reference vs production feature distributions
4. DETECT — PSI > 0.1 on AMT_INCOME_TOTAL triggers DRIFT ALERT: Income distribution shifted
5. ALERT — Slack: "Retraining recommended. Expected default spike: +12%"
6. RECOMMEND — Auto-retrain pipeline or escalate to data scientist for manual review

4. Feature Requirements

4.1 Core Features (Must Have — MVP)

F1: Baseline ML Model

- Train XGBoost on Home Credit Default Risk dataset (Kaggle)
- Features: AMT_INCOME_TOTAL, EXT_SOURCE_1/2/3, AMT_CREDIT, AMT_ANNUITY, DTI ratio
- Output: Default probability score (0.0 - 1.0) per applicant
- Target metric: AUC-ROC >= 0.75 on validation set
- Export model as .pkl file for downstream monitoring

F2: Statistical Drift Detection Engine

- PSI (Population Stability Index) for all numeric features
- Chi-square test for categorical features (OCCUPATION_TYPE, REGION_RATING, etc.)
- Kolmogorov-Smirnov test as secondary numeric drift check
- Drift severity classification: No Drift (PSI<0.1) | Moderate (0.1-0.2) | Severe (>0.2)
- Evidently AI report generation for automated HTML/JSON output
- Drift detection runs as daily scheduled job (configurable)

F3: Alert Engine

- Slack webhook integration — channel-level alert with feature name, PSI score, severity
- Email alert via SMTP with detailed drift report attachment
- Alert deduplication — no repeat alerts within 24h for same feature
- Alert log stored in PostgreSQL with timestamp, feature, score, and severity
- Configurable thresholds per feature (e.g., income PSI > 0.1, EXT_SOURCE PSI > 0.15)

F4: React Dashboard (FastAPI API)

- Overall model health score (composite drift index)
- Feature-level drift table with sortable PSI / KS values
- Historical trend charts — PSI over time per feature
- Distribution comparison plots — reference vs production overlays
- Retraining recommendation panel with one-click retrain trigger
- Alert history log with filter by date/severity

4.2 Extended Features (Should Have — v2)

- Concept drift detection — monitor model output distribution (prediction drift)
- Automated retraining pipeline — trigger model retrain + validation when drift is severe
- Multi-model support — monitor multiple MFI models simultaneously

- REST API — expose drift scores for integration into existing MFI dashboards
- RBI compliance report export (PDF) — monthly audit-ready model monitoring summary

4.3 Future Features (Could Have — v3)

- Explainability layer — SHAP-based feature importance shift analysis
- Anomaly detection on loan officer behavior patterns
- Integration with loan origination systems (LOS) for real-time scoring
- Multi-tenant SaaS mode — onboard multiple MFI clients under one platform

5. Data Requirements

5.1 Dataset

Primary Dataset: Home Credit Default Risk (Kaggle) — application_train.csv (307,511 rows x 122 columns)

File	Rows	Key Columns	Purpose
application_train.csv	307,511	TARGET, AMT_INCOME_TOTAL, EXT_SOURCE_1-3	Model training & drift reference baseline
bureau.csv	1.7M	CREDIT_TYPE, AMT_CREDIT_SUM	Credit bureau history features
installments_payments.csv	13.6M	AMT_PAYMENT, DAYS_INSTALMENT	Repayment behavior features
Synthetic Production Data	Generated	Simulated post-monsoon/inflation shifts	Drift simulation for demo and testing

5.2 Key Features for Drift Monitoring

Feature	Type	Drift Test	Economic Sensitivity
AMT_INCOME_TOTAL	Numeric	PSI + KS	HIGH — Seasonal, inflation-linked
EXT_SOURCE_1/2/3	Numeric	PSI + KS	HIGH — Credit bureau updates
AMT_CREDIT	Numeric	PSI	MEDIUM — Loan demand cycles
DTI Ratio (derived)	Numeric	PSI + KS	HIGH — Income + EMI changes
OCCUPATION_TYPE	Categorical	Chi-square	MEDIUM — Seasonal workforce shifts
REGION_RATING_CLIENT	Categorical	Chi-square	LOW — Stable geographic signal

6. Technical Stack

Layer	Technology	Rationale
ML Model	Python, XGBoost, LightGBM, Scikit-learn	Industry standard for credit risk; 2nd-year feasible
Drift Detection	Evidently AI, SciPy (PSI, KS, Chi-sq)	Purpose-built for ML monitoring; open-source
Dashboard	React, TypeScript, Recharts/Plotly.js	Component-based, typed, interactive web dashboard
Alerts	Slack Webhook API, SMTP (smtplib)	Zero-infra, widely adopted in MFI data teams
Storage	PostgreSQL (drift history, alert logs, model metadata)	Relational storage with transactions and concurrent access for the MVP
Data Pipeline	Pandas, NumPy	Fast data manipulation for batch jobs
Scheduler	APScheduler / cron	Daily monitoring job trigger

7. Non-Functional Requirements

7.1 Performance

- Drift detection job for 10,000 daily loan records: complete within 5 minutes
- Dashboard page load: < 3 seconds
- Alert delivery after detection: < 60 seconds

7.2 Reliability

- Drift job failure must trigger a fallback alert: "Monitoring unavailable — manual review required"
- PostgreSQL data must be backed up daily
- System must handle missing features gracefully (imputation or skip with warning)

7.3 Explainability (RBI Compliance)

- Every alert must include the feature name, statistical test used, PSI/KS score, and severity
- Dashboard must show WHY drift was detected, not just that it was
- Monthly PDF reports must be exportable for regulatory audit

7.4 Scalability

- MVP: Single model, single MFI, PostgreSQL backend; FastAPI serves the dashboard API

- v2: Multi-model, multi-tenant, partner-facing REST API
- v3: Cloud deployment (AWS/GCP), Kubernetes

8. Target Users & Market

8.1 Primary Users

User Type	Pain Point	How DDIS Helps
Data Scientists at MFIs	No visibility into model degradation; discover issues after losses hit	Real-time drift alerts and feature-level diagnostics
Risk Managers	Blind to when ML model scores become unreliable	Dashboard health score; actionable severity ratings
CTO / Head of Product	RBI AI compliance without costly MLOps infra	Audit-ready reports; lightweight open-source stack
Loan Officers	Making decisions on stale model outputs unknowingly	System-level alerts flag when scores are unreliable

8.2 Target Organizations

- Tech-forward MFIs: BharatPe Micro, Paytm Micro, Arohan, Cashpor
- Fintech-NBFCs scaling microloans: KreditBee, Lendingkart, MoneeTap
- 200+ MFIs with ML adoption (50%+ adoption rate in India, 2026)

8.3 Market Size

Metric	Value	Source
Global Data Drift Monitoring Market	$2.38B by 2033 (CAGR 18%)	Industry Research, 2024
India MFI Loan Portfolio	₹3.5 Lakh Cr (2026)	MFIN India Report
MFI ML Adoption (India)	50%+ of mid-to-large MFIs	Industry estimate, 2026
Addressable MFIs	200-300 tech-forward institutions	MFIN / Sa-Dhan data

9. Business Impact & Success Metrics

9.1 Quantified Impact

Metric	Before DDIS	After DDIS (Target)
Default Rate	14%	7-8%
Model Accuracy (AUC)	Drops to 65% undetected	Maintained at 78%+ via timely retrain
Annual Losses	₹15 Cr/year	₹10 Cr/year (-33%)
Drift Detection Time	Weeks/months (manual)	< 24 hours (automated)
Loan Volume	₹100 Cr	₹130 Cr (+30%)
Approval Speed	5 days	24 hours

9.2 KPIs for Product Success

- Time-to-detect drift: < 24 hours from distribution shift occurring
- False positive rate: < 10% (alerts that don't correspond to real drift)
- Dashboard DAU: Data scientists check dashboard >= 3x per week
- Alert-to-action time: Retraining initiated within 48h of severe alert
- MFI satisfaction score: >= 4.2/5 from data scientist users

10. Milestones & Delivery Plan

Phase	Deliverable	Timeline	Owner
Phase 1	Baseline XGBoost model trained + evaluated on Home Credit dataset	Week 1-2	ML Engineer
Phase 2	Drift detection engine — PSI, KS, Chi-square with Evidently integration	Week 3-4	ML/Data Engineer
Phase 3	Alert engine — Slack + Email with PostgreSQL logging and deduplication	Week 5	Backend Dev
Phase 4	React dashboard with drift charts, feature plots, and alert history, backed by FastAPI	Week 6-7	Full Stack / ML

Phase 5	Integration testing, synthetic drift simulation, final demo	Week 8	All

11. Risks & Mitigations

Risk	Probability	Mitigation
Real MFI data not accessible for testing	HIGH	Use synthetic drift simulation on Home Credit data
Evidently AI API changes breaking integration	LOW	Pin to specific Evidently version; build custom PSI fallback
Too many false positive alerts eroding trust	MEDIUM	Tune PSI thresholds per feature; add 24h deduplication
React dashboard and API performance at scale	MEDIUM	Paginate data; cache drift reports; optimize PostgreSQL queries and indexes
RBI regulatory requirements not fully captured	MEDIUM	Consult MFIN/RBI sandbox guidelines; build exportable audit log from day 1

12. Open Questions

- Will pilot MFI partners provide anonymized production data for real-world validation?
- Should auto-retraining be gated behind human approval or fully automated?
- What is the minimum retrain frequency acceptable to MFI risk teams?
- How to handle concept drift (output distribution shift) vs feature drift separately in alerts?
- Should DDIS eventually integrate with RBI's proposed AI model registry?

13. Appendix — Glossary

Term	Definition
Data Drift	Change in the statistical properties of model input features over time compared to training data
PSI	Population Stability Index — measures how much a numeric variable's distribution has shifted; PSI < 0.1 (stable), 0.1-0.2 (moderate), > 0.2 (severe)

KS Test	Kolmogorov-Smirnov test — non-parametric test to compare two distributions
AUC-ROC	Area Under the Receiver Operating Characteristic Curve — primary model quality metric for binary classifiers
DTI	Debt-to-Income ratio — total monthly debt payments divided by gross monthly income
MFI	Microfinance Institution — organizations providing small loans and financial services to low-income individuals
EXT_SOURCE	External credit bureau score features in the Home Credit dataset (normalised 0-1)